QNSMARTI:H:QPCODE

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

COURSE NATURE: DSE (SIGNATURE COURSE)

COURSE CODE: MG4DSEBOTA04

COURSE NAME: VALUE-ADDED PLANT PRODUCTS

(2024 ADMISSION ONWARDS)

Duration: :2 hours **Maximum Marks:70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Define value addition. [K] / [CO1]
2. List any two plant parts commonly used for value addition, with examples [K] / [CO1]
3. Describe bioprospecting [U] / [CO1]
4. Explain vulcanization of rubber [U] / [CO2]
5. Define lyophilization [K] / [CO2]
6. Name the plant from which chocolate is obtained. What is fermentation in chocolate processing? [K] / [CO2]
7. Name any two plants commonly used in cosmetics [K] / [CO3]
8. Define aromatherapy [K] / [CO3]
9. List two advantages of fiber-based craft products. [K] / [CO3]
10. What is branding in plant-based enterprises? [K] / [CO4]

11. Expand FSSAI and AGMARK [U] / [CO4]

12. Name any two government initiatives supporting plant-based startups [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the applications of value addition in food, healthcare, crafts, cosmetics, and bio-materials [U] / [CO1]

14. Discuss traditional and scientific processing techniques used in plant-based products [U] / [CO1]

15. Describe the pre-processing technique [U] / [CO1]

16. Explain various preservation techniques used in value-added plant products [U] / [CO2]

17. Describe the processing and preservation of tea [U] / [CO2]

18. Explain the extraction and formulation of essential oils and hydrosols. [U] / [CO3]

19. Write a note on perfumery plants [A] / [CO3]

20. Explain the cultural significance of eco-crafts in India [U] / [CO3]

21. Explain the role of standards and certification in commercialization of value-added plant products. [U] / [CO4]

22. Describe the challenges faced in commercialization of plant-based value-added products. [U] / [CO4]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23. Explain the principles and steps of value addition. [U] / [CO1]

24. Outline the process of natural dye extraction and application in textile, cosmetics and food [An] / [CO3]

[10x1 = 10]